

Supplementary Information

Supplementary Methods

Contour origin

Spatial-omics contours were generated from Atera WTA cell-coordinate tables and cluster or cell-group definitions. H&E images were not used to draw the breast S1-S5 contours or the cervical tissue contours. Separate H&E segmentation utilities were not used to define the contours analyzed here. H&E whole-slide images entered only after contour generation, through direct-WSI LazySlide PLIP/UNI tile embeddings assigned to registered contour geometry. Atera WTA programs were summarized within the same contour polygons, making each contour the shared unit for image features, molecular features and residual statistical testing. The image-feature and WTA aggregation workflows are described in Online Methods; implementation details are available to editors through the editorial submission materials.

Non-independence of contour construction and residual estimand

The spatial-omics origin of the contours means that they are not independent histological labels. The analysis therefore estimates within-label residual association rather than independent contour discovery. We denote the spatial-omics-derived discrete contour label as C , spatial covariates including centroid and boundary-distance bins as S , the continuous WTA program score as Y and the H&E foundation-model feature as X . The reported association is computed between residualized $X / C, S$ and residualized $Y / C, S$. This does not prove standalone morphology-only clinical prediction and does not exclude all contour-construction-induced coupling, but it argues against the simpler explanation that mTM merely rediscovers the discrete spatial-omics contour label.

23 Symbol table for the residual model

Symbol	Meaning
i	contour index
C_i	spatial-omics-derived discrete contour label
S_i	spatial covariates, including centroid coordinates and boundary-distance bins
X_{ik}	H&E foundation-model feature k in contour i
Y_i	continuous WTA program score in contour i
Z_i	covariate design vector $[1, \text{onehot}(C_i), S_i]$
r_k^X	residualized rank-transformed H&E feature
r^Y	residualized rank-transformed WTA program score
ρ_k	residual partial Spearman association between r_k^X and r^Y

24 Residual statistical model

25 Contour-level molecular and image features were rank transformed and residualized against
 26 specified covariates: spatial-omics-derived contour label, centroid x/y coordinates and available
 27 boundary-distance summaries. For the breast S3 analysis, the contour label was constant and was
 28 omitted from the model matrix. In matrix notation,

$$29 \quad Z_i = [1, \text{onehot}(C_i), S_i],$$

$$30 \quad r_k^X = (I - P_Z) \text{rank}(X_k), r^Y = (I - P_Z) \text{rank}(Y),$$

$$31 \quad \rho_k = \text{cor}(r_k^X, r^Y), P_Z = Z(Z^T Z)^{-1} Z^T.$$

32 Associations were reported as partial Spearman's ρ between residualized image and WTA
 33 program vectors.

34 Spatial permutation and block bootstrap

35 For the spatial-null defense, molecular residuals were permuted within strata preserving contour
 36 label, centroid-position bins and boundary-distance bins. This retains coarse compartmental and
 37 spatial organization while breaking contour-wise H&E-to-WTA pairing. Empirical P values
 38 were calculated as

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$$P_{emp} = \frac{1 + \sum_b 1\{|\rho_b^{null}| \geq |\rho_{obs}|\}}{1 + B}.$$

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57 **Boundary co-variation analysis**

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The permutation mitigates coarse spatial-autocorrelation explanations but does not exclude all fine-scale spatial dependence, registration uncertainty or biological coupling induced by the original spatial-omics contour construction. Spatial block bootstrap resampled centroid x/y quartile blocks to estimate confidence intervals.

Additional sensitivity checks were run after the locked source-data package was assembled.

Candidate associations were recomputed after leaving out each of 16 spatial blocks, compared with 1,000 local mismatched-pair controls and recalculated under centroid-covariate jitter up to

1% of slide span. Existing tile embeddings were also reassigned under 23 small registration

perturbations, including translations, rotations, scale changes and mild combined shifts, before

recomputing candidate contour-level embedding means and residual associations. In a nested

spatial holdout check, embedding features were selected using only training spatial blocks and

evaluated on held-out blocks. Component-gene audits recomputed contour-level means for genes

inside the selected breast S3 WTA programs and tested whether the reported H&E embedding

axis tracked these component genes under the same residual framework. Cervical claims are

based on embedding-derived WTA program associations; PLIP text-similarity outputs from the

breast-oriented prompt set were not used as cervical biological claims. These checks strengthen

transcript-level and statistical consistency but do not replace independent IHC/protein validation.

Boundary profiles were computed in distance rings around selected tissue interfaces. Candidate

molecularly active zones were interpreted only as ring-level H&E-WTA co-variation. Causal,

temporal or directional boundary interpretations were not assigned.

Figure source-data and supplementary data mapping

- `Figure_1b_Hero_Patches_Source_Data.csv`: contour IDs, WTA z-scores, oriented H&E embedding z-scores and contour-size metadata for QC-filtered breast S3 hero patches.
- `Figure_1c_Spatial_Permutation_Source_Data.csv`: spatial-null permutation results for reported candidate programs.
- `Figure_1c_BlockBootstrap_Source_Data.csv`: spatial block-bootstrap confidence intervals.
- `Figure_1d_MAZ_QC_Source_Data.csv`: conservative boundary co-variation summaries used for the MAZ panel and Supplementary Fig. 3.
- `Figure_1e_CrossCancer_Signature_Source_Data.csv`: program-family signature summary across breast/cervical and PLIP/UNI.

The five files above are figure source data for Fig. 1 and Supplementary Figs. 1-3.

Supplementary Tables 1-4 reuse those same CSVs because they tabulate the spatial permutation, bootstrap, cross-cancer signature and hero-patch values shown in Fig. 1.

Separate machine-readable Supplementary Data CSVs accompany Supplementary Tables 5-9:

- `Supplementary_Table_5_SpatialSensitivity.csv`: leave-one-block, local mismatch and centroid-jitter sensitivity summary.
- `Supplementary_Table_6_GeneComponent_Summary.csv`: breast S3 PLIP component-gene audit summary.
- `Supplementary_Table_6_GeneComponent_Long.csv`: per-gene component audit values for selected breast S3 PLIP programs.

- Supplementary_Table_7_RegistrationPerturbation_Summary.csv: registration-perturbation summary by candidate.
- Supplementary_Table_7_RegistrationPerturbation_Long.csv: per-perturbation registration-sensitivity values.
- Supplementary_Table_8_NestedSpatialHoldout_Summary.csv: nested spatial holdout summary by candidate.
- Supplementary_Table_8_NestedSpatialHoldout_Long.csv: per-fold nested holdout values.
- Supplementary_Table_9_ContourSizeSensitivity.csv: contour-size sensitivity for breast and cervical PLIP/UNI candidate associations.

Supplementary Tables

Supplementary Table 1. Spatial permutation defense

Dataset	Model	Program	Program family	n contours	Observed partial Spearman's ρ	Permutations	Empirical P	99% null threshold
Breast	PLIP	luminal_estrogen_response	endocrine/epithelial identity	157	-0.639	10,000	9.999e-5	0.306
Breast	PLIP	unfolded_protein_response	metabolic/stress	157	0.515	10,000	9.999e-5	0.232
Breast	PLIP	oxidative_phosphorylation	metabolic/stress	157	0.531	10,000	9.999e-5	0.290
Cervical	PLIP	myofibroblast_caf_activation	stromal-remodeling/CAF/ECM	215	-0.555	10,000	9.999e-5	0.450
Cervical	PLIP	emt_invasive_front	invasion/boundary/EMT	215	0.552	10,000	9.999e-5	0.430
Cervical	PLIP	tls_adjacent_activation	immune ecology/TLS/immune exclusion	215	0.551	10,000	9.999e-5	0.465
Cervical	PLIP	collagen_ecm_organization	stromal-remodeling/CAF/ECM	215	-0.546	10,000	9.999e-5	0.461
Cervical	PLIP	immune_exclusion	immune ecology/TLS/immune exclusion	215	0.544	10,000	9.999e-5	0.446
Cervical	PLIP	stromal_encapsulation	stromal-remodeling/CAF/ECM	215	0.541	10,000	9.999e-5	0.461
Cervical	PLIP	epithelial_identity	endocrine/epithelial identity	215	0.522	10,000	0.0023	0.497
Cervical	PLIP	immune_activation	immune ecology/TLS/immune exclusion	215	0.505	10,000	9.999e-5	0.448

Dataset	Model	Program	Program family	n contours	Observed partial Spearman's ρ	Permutations	Empirical P	99% null threshold
Cervical	PLIP	oxidative_phosphorylation	immune exclusion metabolic/stress	215	-0.487	10,000	9.999e-5	0.396
Cervical	PLIP	emt_invasion	invasion/ boundary/ EMT	215	0.479	10,000	9.999e-5	0.407

95 Supplementary Table 2. Spatial block-bootstrap confidence intervals

Dataset	Model	Program	Bootstrap samples	Median ρ	95% CI
Breast	PLIP	luminal_estrogen_response	2,000	-0.652	-0.742 to -0.528
Breast	PLIP	unfolded_protein_response	2,000	0.519	0.358 to 0.658
Breast	PLIP	oxidative_phosphorylation	2,000	0.531	0.404 to 0.657
Cervical	PLIP	myofibroblast_caf_activation	2,000	-0.553	-0.628 to -0.484
Cervical	PLIP	emt_invasive_front	2,000	0.552	0.425 to 0.646
Cervical	PLIP	tls_adjacent_activation	2,000	0.551	0.475 to 0.631
Cervical	PLIP	collagen_ecm_organization	2,000	-0.543	-0.624 to -0.469
Cervical	PLIP	immune_exclusion	2,000	0.544	0.431 to 0.656
Cervical	PLIP	stromal_encapsulation	2,000	0.539	0.440 to 0.628
Cervical	PLIP	epithelial_identity	2,000	0.524	0.386 to 0.616
Cervical	PLIP	immune_activation	2,000	0.504	0.405 to 0.597
Cervical	PLIP	oxidative_phosphorylation	2,000	-0.488	-0.631 to -0.302
Cervical	PLIP	emt_invasion	2,000	0.482	0.404 to 0.558

96 Supplementary Table 3. Program-family signatures by cancer and model

Dataset	Program family	Model	Maximum absolute partial Spearman's ρ	Top WTA program	Support
Breast	endocrine/epithelial identity	PLIP	0.597	luminal_estrogen_response	strong
Breast	endocrine/epithelial identity	UNI	0.586	luminal_estrogen_response	strong
Breast	metabolic/stress	PLIP	0.491	unfolded_protein_response	strong
Breast	metabolic/stress	UNI	0.456	oxidative_phosphorylation	strong
Breast	stromal-remodeling/CAF/ECM	PLIP	0.330	collagen_ecm_organization	weak
Breast	stromal-remodeling/CAF/ECM	UNI	0.377	collagen_ecm_organization	moderate
Breast	immune ecology/TLS/immune exclusion	PLIP	0.409	tls_b_cell_plasma	moderate
Breast	immune ecology/TLS/immune exclusion	UNI	0.391	t_cell_exhaustion_checkpoint	moderate

Dataset	Program family	Model	Maximum absolute partial Spearman's ρ			Support
			Top WTA program			
Breast	invasion/boundary/EMT	PLIP	NA	not_detected		not_detected
Breast	invasion/boundary/EMT	UNI	0.304	emt_invasion		weak
Cervical	endocrine/epithelial identity	PLIP	0.522	epithelial_identity		strong
Cervical	endocrine/epithelial identity	UNI	0.527	epithelial_identity		strong
Cervical	metabolic/stress	PLIP	0.487	oxidative_phosphorylation		strong
Cervical	metabolic/stress	UNI	0.555	oxidative_phosphorylation		strong
Cervical	stromal-remodeling/CAF/ECM	PLIP	0.555	myofibroblast_caf_activation		strong
Cervical	stromal-remodeling/CAF/ECM	UNI	0.615	stromal_encapsulation		strong
Cervical	immune ecology/TLS/immune exclusion	PLIP	0.551	tls_adjacent_activation		strong
Cervical	immune ecology/TLS/immune exclusion	UNI	0.633	immune_exclusion		strong
Cervical	invasion/boundary/EMT	PLIP	0.552	emt_invasive_front		strong
Cervical	invasion/boundary/EMT	UNI	0.534	emt_invasion		strong

97 Supplementary Table 4. Final luminal estrogen-response hero patch pairs

Pair	Polarity	Contour ID	WTA program z-score	Oriented H&E embedding z-score	n_tiles	n_cells
1	high	S3 S3 #417.1	1.03	0.78	7	508
1	low	S3 S3 #519.1	-0.82	-1.06	3	211
2	high	S3 S3 #139.1	0.52	1.28	6	328
2	low	S3 S3 #56.1	-0.68	-0.59	6	120
3	high	S3 S3 #351.1	0.82	0.92	10	138
3	low	S3 S3 #188.1	-0.83	-0.93	6	59
4	high	S3 S3 #173.1	1.74	0.43	5	566
4	low	S3 S3 #285.1	-0.61	-0.84	3	78

98 Supplementary Table 5. Spatial sensitivity summary

Dataset	Model	Candidate programs	Leave-one-block sign-stable	Exceed local mismatch q95	Minimum centroid-jitter sign stability
Breast	PLIP	3	3/3	3/3	1.00
Breast	UNI	5	5/5	5/5	1.00
Cervical	PLIP	10	10/10	10/10	1.00
Cervical	UNI	5	5/5	5/5	1.00

99 Supplementary Table 6. Breast S3 component-gene audit

Program	Program genes found	Effective image-gene n	Reported program-image ρ	Median image-gene ρ	Image-gene sign-match fraction	Program high-low positive fraction	Strongest component gene
luminal_estrogen_response	9	157	-0.639	-0.560	1.00	1.00	XBP1

Program	Program genes found	Effective image-gene n	Reported program- image ρ	Median image-gene ρ	Image-gene sign- match fraction	Program high-low positive fraction	Strongest component gene
unfolded_protein_response	9	157	0.515	0.285	0.78	1.00	XBP1
oxidative_phosphorylation	8	157	0.531	0.341	0.88	1.00	COX6C

100 **Supplementary Table 7. Registration perturbation sensitivity**

Dataset	Model	Program	Perturbations	Base ρ	Median perturbed ρ	Max abs delta ρ	Sign-stable fraction	Min assignment fraction
Breast	PLIP	luminal_estrogen_response	23	-0.652	-0.652	0.054	1.000	0.999
Breast	PLIP	unfolded_protein_response	23	0.523	0.520	0.038	1.000	0.999
Breast	PLIP	oxidative_phosphorylation	23	0.536	0.536	0.041	1.000	0.999
Cervical	PLIP	myofibroblast_caf_activation	23	-0.555	-0.548	0.077	1.000	0.999
Cervical	PLIP	emt_invasive_front	23	0.552	0.552	0.031	1.000	0.999
Cervical	PLIP	tls_adjacent_activation	23	0.551	0.546	0.034	1.000	0.999
Cervical	PLIP	collagen_ecm_organization	23	-0.546	-0.543	0.066	1.000	0.999
Cervical	PLIP	immune_exclusion	23	0.544	0.534	0.062	1.000	0.999
Cervical	PLIP	stromal_encapsulation	23	0.541	0.535	0.060	1.000	0.999
Cervical	PLIP	epithelial_identity	23	0.522	0.509	0.137	1.000	0.999
Cervical	PLIP	immune_activation	23	0.505	0.519	0.045	1.000	0.999
Cervical	PLIP	oxidative_phosphorylation	23	-0.487	-0.492	0.071	1.000	0.999
Cervical	PLIP	emt_invasion	23	0.479	0.465	0.070	1.000	0.999

101 **Supplementary Table 8. Nested spatial holdout**

Dataset	Model	Program	Folds	Base locked ρ	Median held-out selected ρ	Selected sign- match fraction	Locked sign- match fraction	Locked-feature reuse fraction
Breast	PLIP	luminal_estrogen_response	16	-0.652	-0.637	0.938	0.938	1.000
Breast	PLIP	unfolded_protein_response	16	0.523	0.462	0.750	0.750	1.000
Breast	PLIP	oxidative_phosphorylation	16	0.536	0.367	0.750	0.812	0.000
Cervical	PLIP	myofibroblast_caf_activation	16	-0.555	-0.486	0.750	0.812	0.875
Cervical	PLIP	emt_invasive_front	16	0.552	0.626	0.750	0.750	1.000
Cervical	PLIP	tls_adjacent_activation	16	0.551	0.516	0.625	0.625	0.688
Cervical	PLIP	collagen_ecm_organization	16	-0.546	-0.531	0.562	0.625	0.750
Cervical	PLIP	immune_exclusion	16	0.544	0.593	0.688	0.688	0.938
Cervical	PLIP	stromal_encapsulation	16	0.541	0.430	0.750	0.750	0.875
Cervical	PLIP	epithelial_identity	16	0.522	0.395	0.688	0.688	1.000
Cervical	PLIP	immune_activation	16	0.505	0.383	0.562	0.562	0.188
Cervical	PLIP	oxidative_phosphorylation	16	-0.487	-0.188	0.562	0.688	0.750
Cervical	PLIP	emt_invasion	16	0.479	0.729	0.875	0.875	0.875
Breast	UNI	luminal_estrogen_response	16	-0.586	-0.422	0.812	0.938	0.812

Dataset	Model	Program	Folds	Base locked ρ	Median held-out selected ρ	Selected sign-match fraction	Locked sign-match fraction	Locked-feature reuse fraction
Breast	UNI	oxidative_phosphorylation	16	-0.456	-0.493	0.938	0.938	0.938
Breast	UNI	collagen_ecm_organization	16	-0.377	-0.259	0.750	0.750	1.000
Breast	UNI	t_cell_exhaustion_checkpoint	16	0.391	0.442	0.875	0.875	1.000
Breast	UNI	emt_invasion	16	0.304	0.157	0.750	0.750	1.000
Cervical	UNI	epithelial_identity	16	-0.527	-0.288	0.688	0.688	0.938
Cervical	UNI	oxidative_phosphorylation	16	0.555	0.562	0.625	0.625	0.688
Cervical	UNI	stromal_encapsulation	16	-0.615	-0.412	0.688	0.750	0.750
Cervical	UNI	immune_exclusion	16	-0.633	-0.631	0.750	0.750	1.000
Cervical	UNI	emt_invasion	16	0.534	-0.146	0.375	0.688	0.500

Supplementary Table 9. Contour-size sensitivity

The machine-readable table contains full breast and cervical PLIP/UNI candidate results under the full set, n_tiles-only, n_cells-only and combined size filters. The compact view below shows the primary breast S3 PLIP programs under the full set and the combined n_tiles ≥ 3 and n_cells ≥ 50 filter.

Program	Filter	n contours	Recomputed partial Spearman's rho
luminal_estrogen_response	full set	157	-0.652
luminal_estrogen_response	n_tiles ≥ 3 ; n_cells ≥ 50	49	-0.656
unfolded_protein_response	full set	157	0.523
unfolded_protein_response	n_tiles ≥ 3 ; n_cells ≥ 50	49	0.672
oxidative_phosphorylation	full set	157	0.536
oxidative_phosphorylation	n_tiles ≥ 3 ; n_cells ≥ 50	49	0.647

Supplementary Figure Captions

Supplementary Fig. 1 | Spatial permutation defense. Program residuals were permuted within compartment-aware strata defined by spatial-omics-derived contour labels, centroid-position bins and boundary-distance bins. **A**, observed residual associations after structure/spatial residualization. **B**, observed absolute partial correlations compared with the 95th percentile of the stratified spatial null. This test mitigates coarse spatial-autocorrelation explanations but does not exclude all fine-scale spatial dependence.

Supplementary Fig. 2 | Expanded breast S3 luminal estrogen-response examples. High- and low-program S3 contours are shown as expanded H&E examples for Fig. 1b. All displayed

patches pass $n_tiles \geq 3$, $n_cells \geq 50$, edge-proximity and area QC filters. They share the same spatial-omics-derived contour label and were selected as representative concordant examples after statistical ranking; they are not a blinded visual diagnostic claim.

Supplementary Fig. 3 | Expanded boundary co-variation profiles. Ring-level profiles show H&E-WTA co-variation at selected tissue interfaces. These examples illustrate conservative boundary co-variation and do not imply causality, temporal ordering or directional boundary effects.

Source Data

Figure Source Data files accompany Fig. 1 and Supplementary Figs. 1-3. Separate Supplementary Data CSVs accompany Supplementary Tables 5-9 so that robustness values can be reviewed in machine-readable form rather than only through PDF table rendering. No IHC or protein validation is claimed.